BoT-IoT MAQT Training Setups

This note summarizes how **lawun** chose MAQT training settings on FROZEN BoT-IoT, and how those runs compare to **arjun**'s reported CE-head test accuracy.

Unless stated otherwise, confusion matrices below are **Algo-3 accepted-known test** results: known-class test rows with nonconformity $s(x) \leq q$ (Global conformal). Rejected known-test rows are excluded from accuracy and from the matrix.

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1. Baseline: 4-class, 3000 / class, fixed LR 0.05

Setup: four known classes (DDoS, DoS, Normal, Reconnaissance); DPP coreset capped at **3000 per class**; fixed learning rate 0.05.

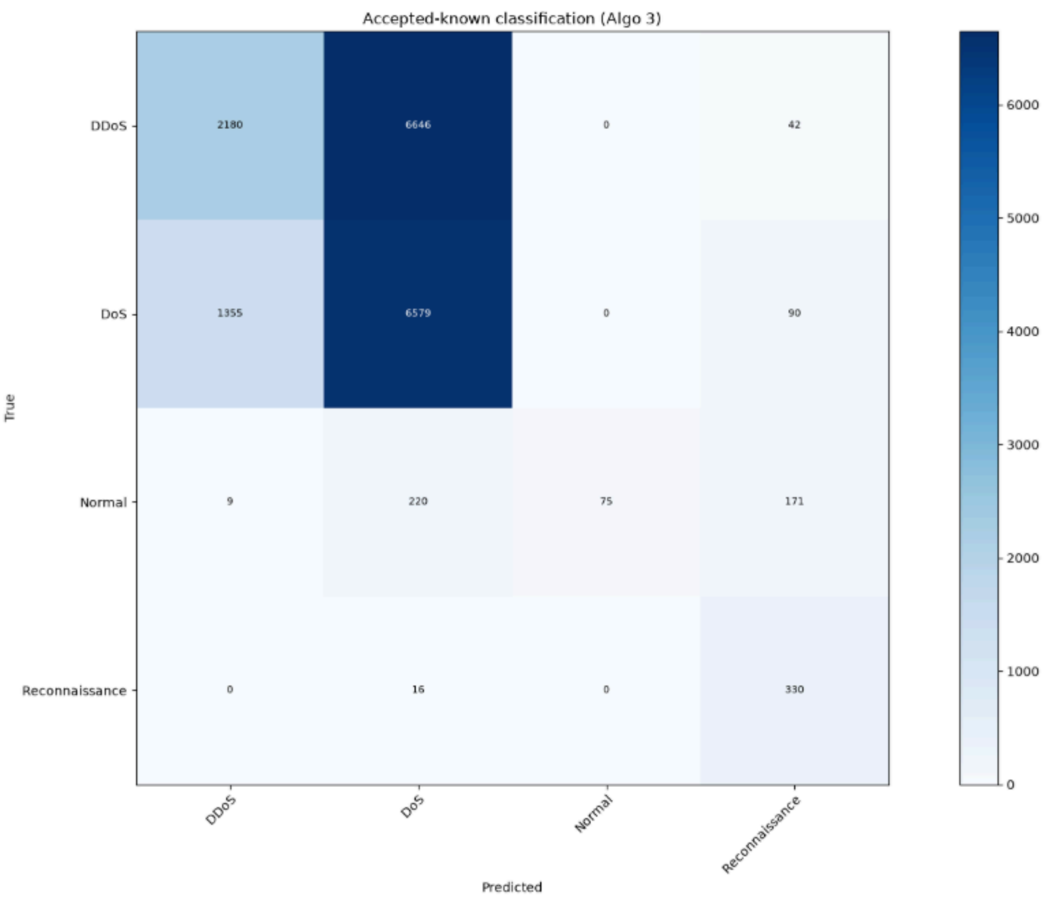
Observation: overall accepted accuracy is modest (~52%). **Normal** is poorly recovered (high precision, very low recall). **DDoS** and **DoS** are heavily confused.

```
[test] mean certified radius (accepted): 0.0125
[test] reject rate (known): 0.0472 | acc (accepted): 0.5174

[test] classification report (accepted only):
                precision    recall  f1-score   support

          DDoS       0.62      0.25      0.35      8868
           DoS       0.49      0.82      0.61      8024
        Normal       1.00      0.16      0.27       475
Reconnaissance       0.52      0.95      0.67       346

      accuracy                           0.52     17713
     macro avg       0.66      0.54      0.48     17713
  weighted avg       0.57      0.52      0.47     17713
```



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2. Ablation: drop DoS (3-class), same budget

Hypothesis: majority DoS / DDoS traffic might be starving **Normal**.

Setup: train without **DoS** (DDoS + Normal + Reconnaissance only); still **3000 / class**, LR 0.05.

Observation: Normal does **not** improve (accepted Normal recall stays at zero in this run). High headline accuracy is driven by DDoS dominance, not better Normal geometry.

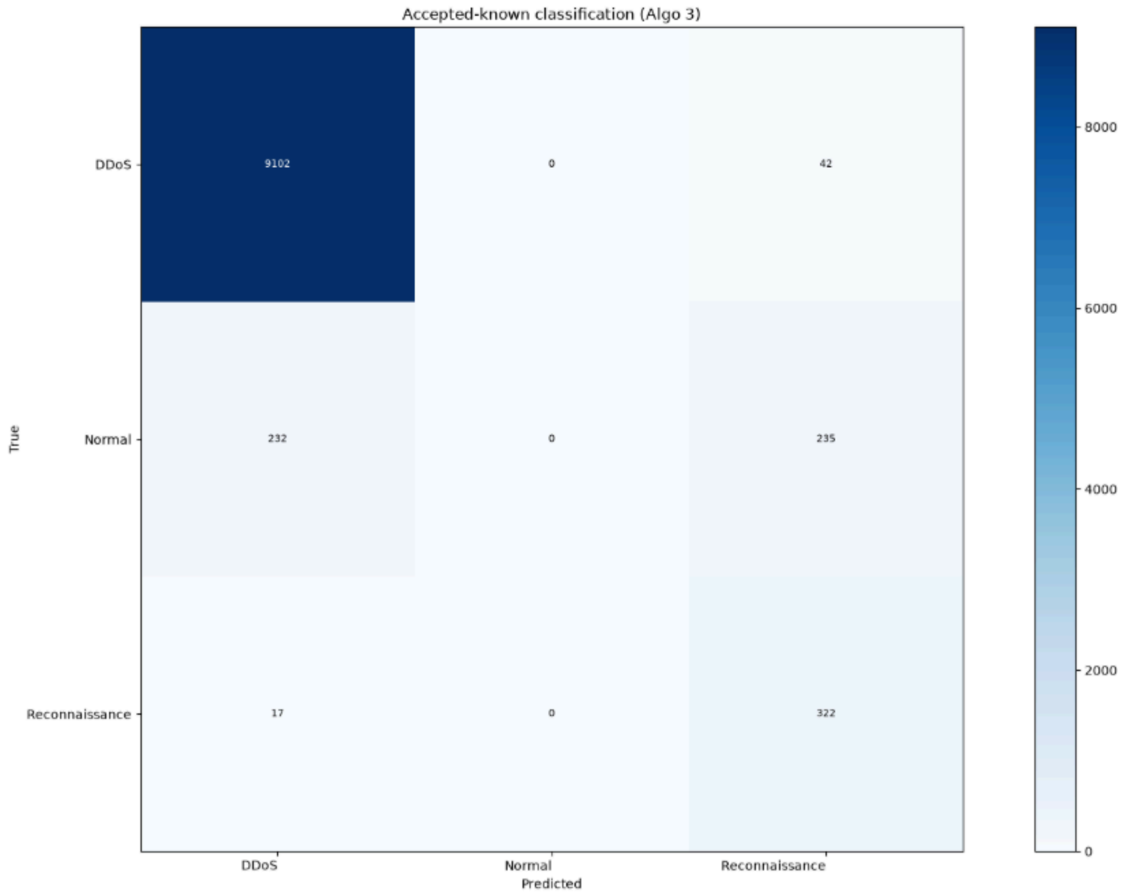
Conclusion: DoS presence is not the main cause of poor Normal prediction $\longrightarrow$ return to **4-class**.

```
[test] mean certified radius (accepted): 0.0710
[test] reject rate (known): 0.0484 | acc (accepted): 0.9471

[test] classification report (accepted only):
                precision    recall  f1-score   support

          DDoS       0.97      1.00      0.98      9144
        Normal       0.00      0.00      0.00       467
Reconnaissance       0.54      0.95      0.69       339

      accuracy                           0.95      9950
     macro avg       0.50      0.65      0.56      9950
  weighted avg       0.91      0.95      0.93      9950
```



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3. Scale data: 4-class, 20000 / class, LR 0.05

Setup: larger DPP coreset (**20000 per class**), same fixed LR 0.05.

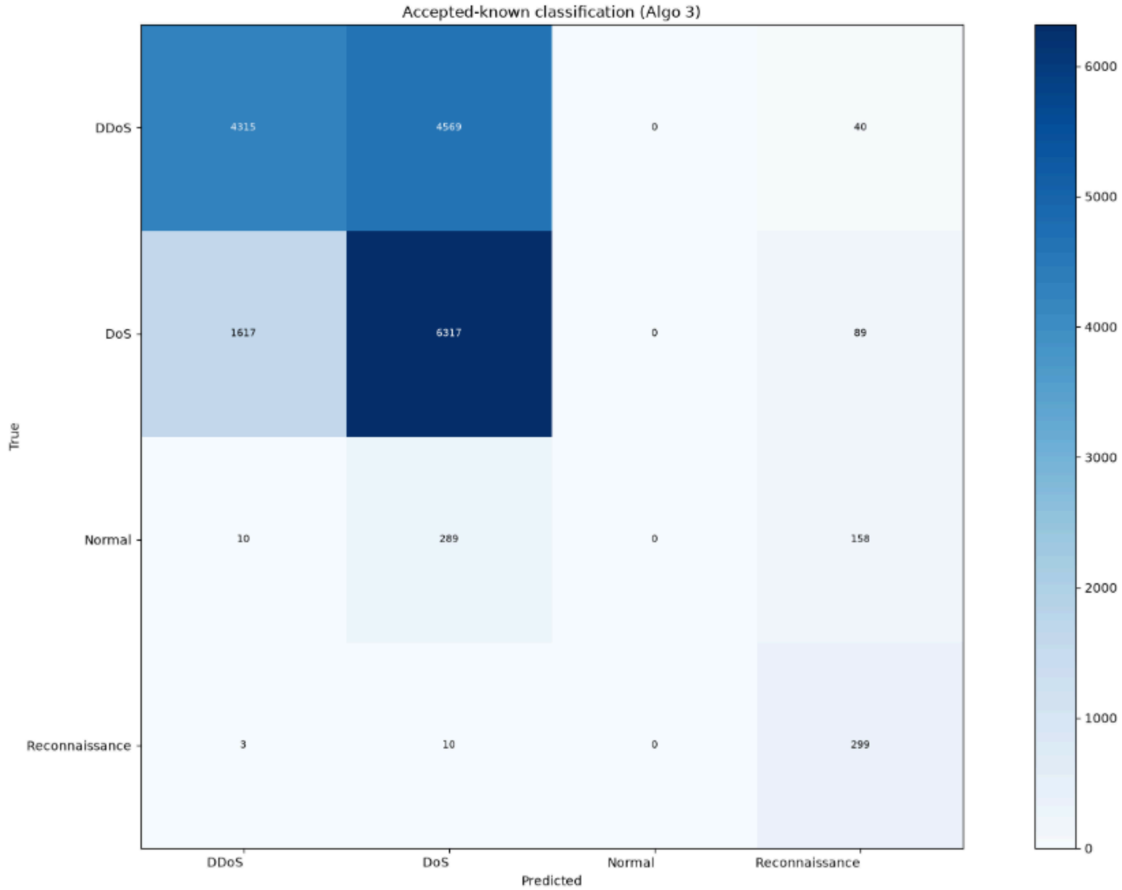
Observation: accepted accuracy rises (~62% vs ~52% at 3000). **DDoS** $\leftrightarrow$ **DoS** confusion remains large; **Normal** stays weak (near-zero correct in this matrix). More data helps overall accuracy but does not fix class separation.

```
[test] mean certified radius (accepted): 0.0082
[test] reject rate (known): 0.0471 | acc (accepted): 0.6170

[test] classification report (accepted only):
                precision    recall  f1-score   support

          DDoS       0.73      0.48      0.58      8924
           DoS       0.56      0.79      0.66      8023
        Normal       0.00      0.00      0.00       457
Reconnaissance       0.51      0.96      0.67       312

      accuracy                           0.62     17716
     macro avg       0.45      0.56      0.48     17716
  weighted avg       0.63      0.62      0.60     17716
```



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4. Smaller fixed LR, moderate size: 10000 / class, LR 0.0005

Setup: 10000 per class, fixed LR 0.0005.

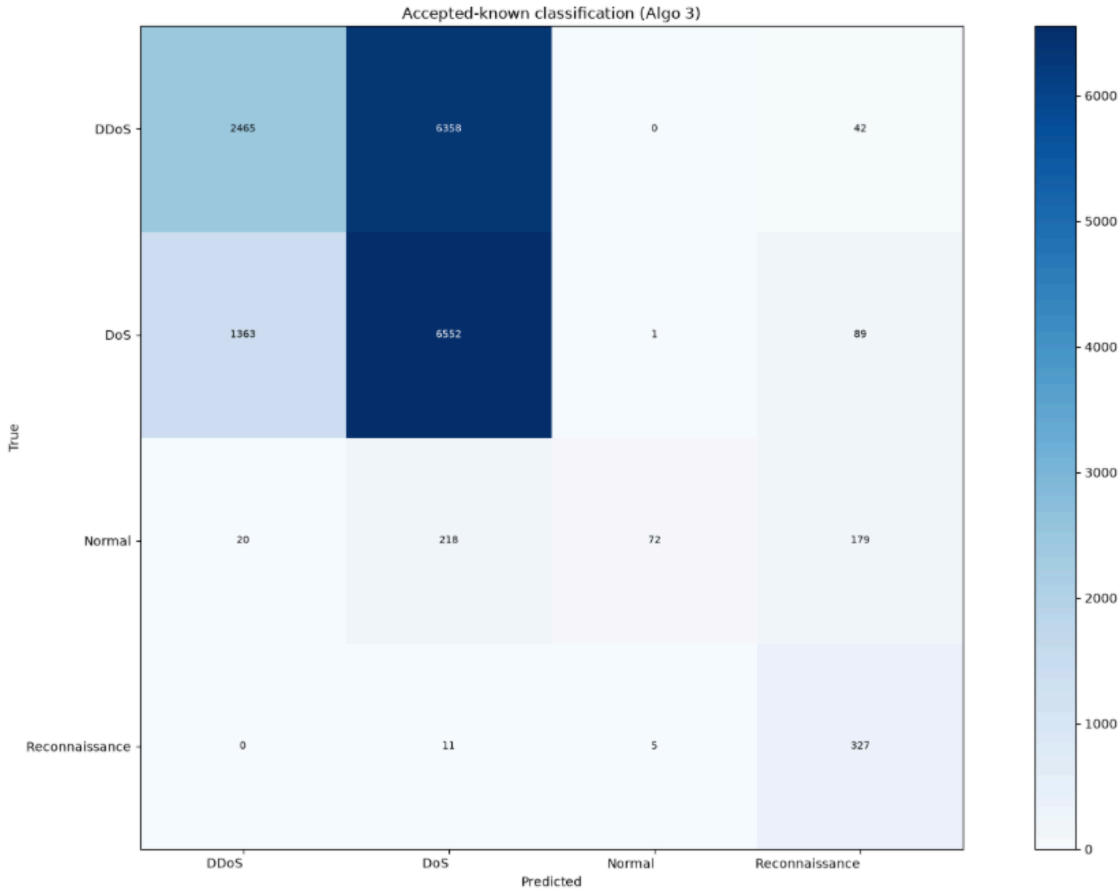
Observation: accepted accuracy (~53%) is close to the 3000 / LR 0.05 baseline (only a small bump, on the order of ~1Auto LR percentage point in this comparison). **DDoS** ↔ **DoS** and weak **Normal** persist. Extra samples + tiny fixed LR do not justify the cost for this pipeline.

```
[test] mean certified radius (accepted): 0.0132
[test] reject rate (known): 0.0478 | acc (accepted): 0.5319

[test] classification report (accepted only):
                precision    recall  f1-score   support

          DDoS       0.64      0.28      0.39      8865
           DoS       0.50      0.82      0.62      8005
        Normal       0.92      0.15      0.25       489
Reconnaissance       0.51      0.95      0.67       343

      accuracy                           0.53     17702
     macro avg       0.64      0.55      0.48     17702
  weighted avg       0.58      0.53      0.49     17702
```



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5. Back to 3000 / class with cosine LR schedule

Given diminishing returns from larger coresets under fixed LR, lawun returns to **3000 per class** and turns on **automatic LR decay** (`use_lr_schedule=True`, cosine annealing from the starting LR toward a small floor).

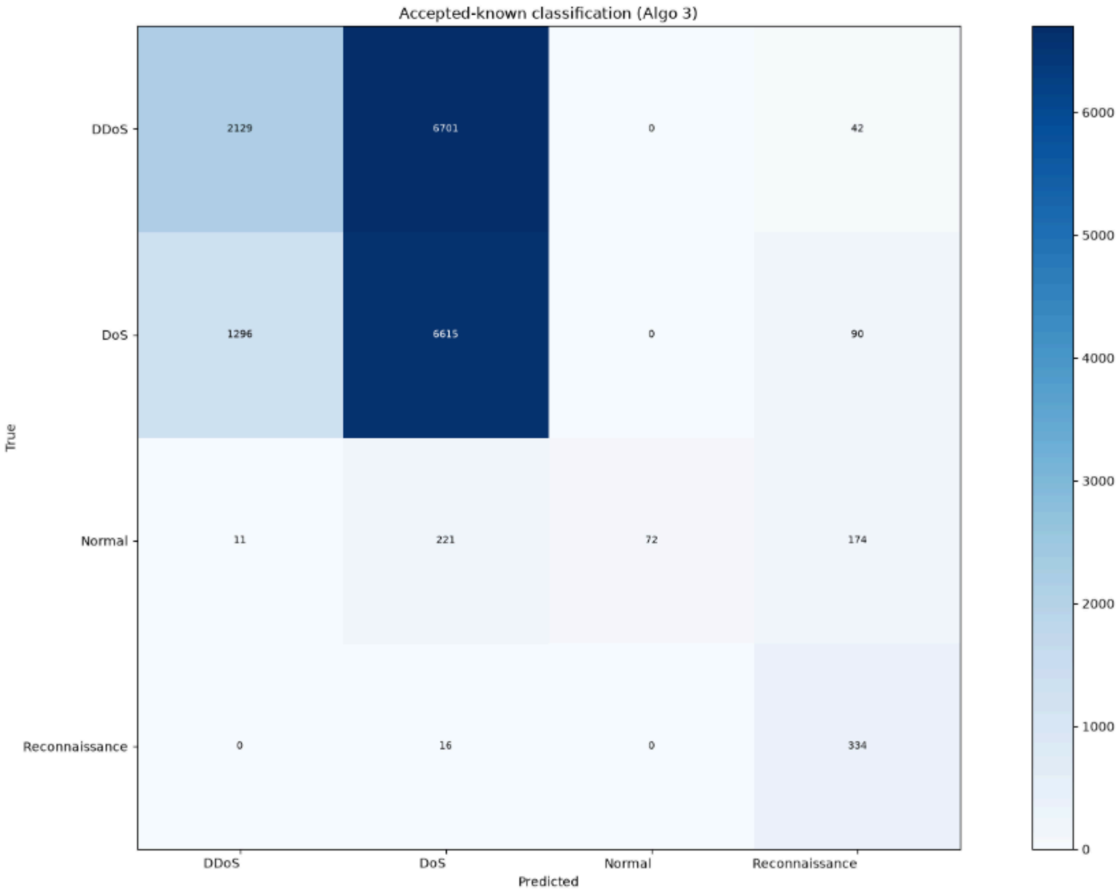
5a. Start LR 0.03

```
[test] mean certified radius (accepted): 0.0136
[test] reject rate (known): 0.0479 | acc (accepted): 0.5169

[test] classification report (accepted only):
                precision    recall  f1-score   support

          DDoS       0.62      0.24      0.35      8872
           DoS       0.49      0.83      0.61      8001
        Normal       1.00      0.15      0.26       478
Reconnaissance       0.52      0.95      0.67       350

      accuracy                           0.52     17701
     macro avg       0.66      0.54      0.47     17701
  weighted avg       0.57      0.52      0.47     17701
```



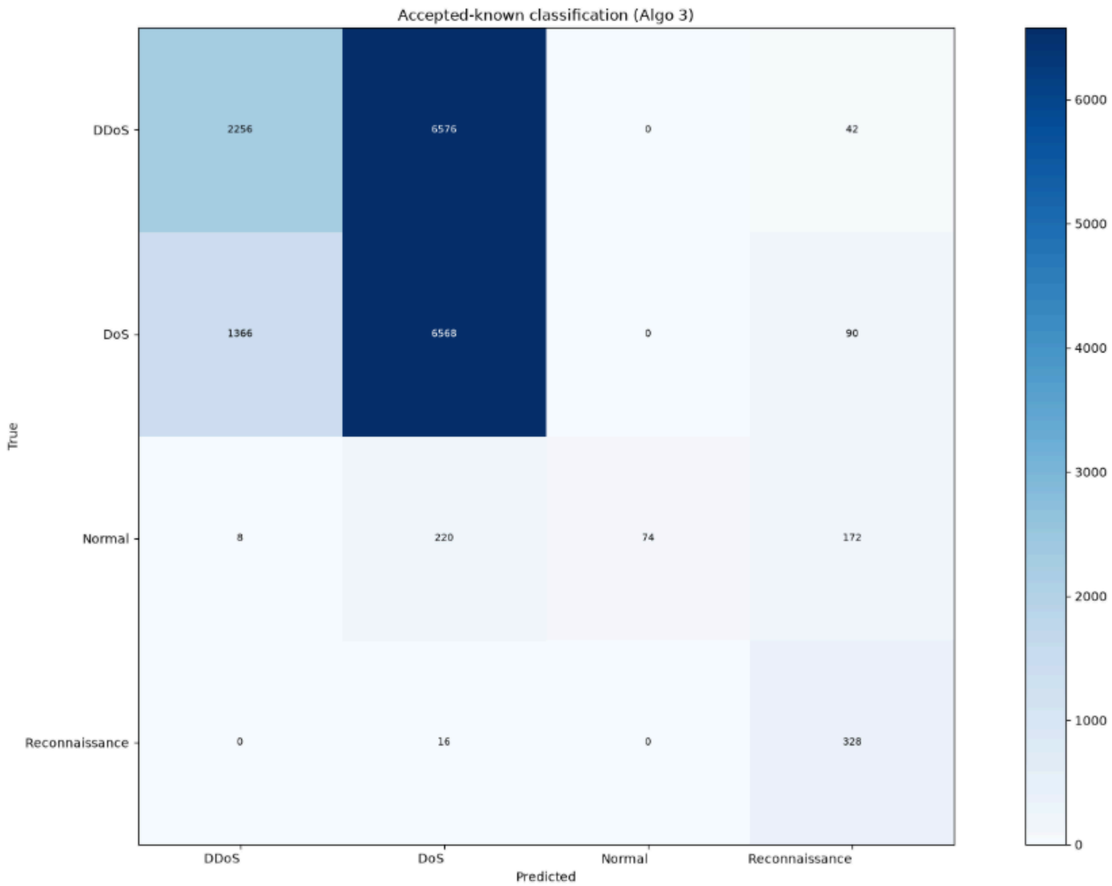
5b. Start LR 0.01 (chosen)

```
[test] mean certified radius (accepted): 0.0120
[test] reject rate (known): 0.0471 | acc (accepted): 0.5208

[test] classification report (accepted only):
                precision    recall  f1-score   support

          DDoS       0.62      0.25      0.36      8874
           DoS       0.49      0.82      0.61      8024
        Normal       1.00      0.16      0.27       474
Reconnaissance       0.52      0.95      0.67       344

      accuracy                           0.52     17716
     macro avg       0.66      0.55      0.48     17716
  weighted avg       0.57      0.52      0.48     17716
```



Observation: accepted metrics for start 0.03 vs 0.01 are similar (~52%). Training with start 0.01 is judged smoother on the loss/LR curves, so the working choice is:

Knob	Choice
Classes	4 (DDoS, DoS, Normal, Reconnaissance)
Train coreset	3000 / class (DPP)
LR	cosine schedule, start 0.01
Notebooks	`4class-3000each-lr0.01auto-maqt` $\to$ `4class-3000each-lr0.01auto-algo3` (+ cached variants)

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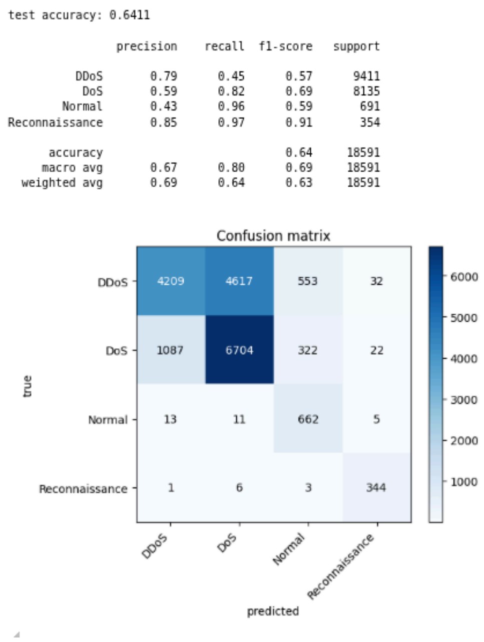
6. Fair compare to arjun (full known test, no reject mask)

Algo-3 accepted-only accuracy is **not** comparable to a CE-head score on **every** known-test row.

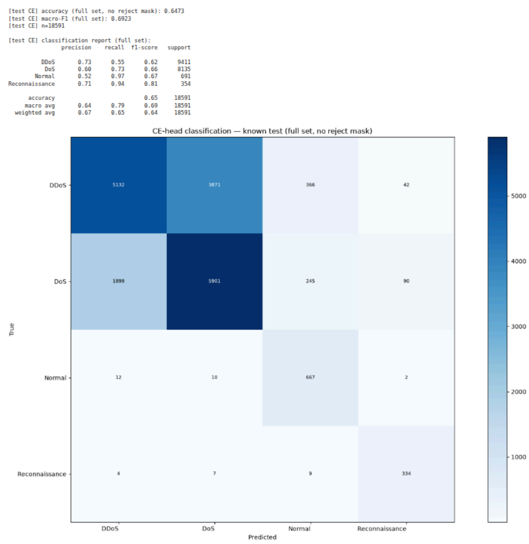
- **arjun** (`ablation-test-bot-iot`): CE-head `predict_labels` on full known test $\to$ about **64%** accuracy ($n = 18591$).

- **lawun** best MAQT (4-class, 3000 / class, cosine start 0.01): CE-head on the **same full known test**, no conformal mask $\longrightarrow$ about **65%** accuracy ($n = 18591$).

arjun (~64%, full known test, CE, no reject)



lawun (~65%, full known test, CE, no reject)



Under that matched protocol, lawun's selected run is on par with (slightly above) arjun's reported CE full-test accuracy. Remaining open issue for both setups: **DDoS vs DoS** separation; conformal accepted-only CMs still show weak **Normal** recall when rejects are stripped out.

Decision trail (short)

```
4c / 3k / lr=0.05
  $\rightarrow$ Normal weak, DDoS↔DoS confused
3c (drop DoS) / 3k / lr=0.05
  $\rightarrow$ Normal still not fixed $\rightarrow$ back to 4c
4c / 20k / lr=0.05
  $\rightarrow$ accuracy up; DDoS↔DoS / Normal still hard
4c / 10k / lr=0.0005
  $\rightarrow$ ≈ 3k result; not worth the extra data for this goal
4c / 3k / cosine 0.03 vs 0.01
  $\rightarrow$ similar accepted metrics; prefer 0.01 (smoother)
selected: 4c / 3k / cosine start 0.01
fair CE full-test vs arjun: ~65% vs ~64%
```